Mahindra University Hyderabad

École Centrale School of Engineering
Minor-II Examination

Program: B. Tech. Branch: All Year: 1st Semester: 1st
Subject: Introduction to Computing (CS1001)
All Branches

Date: 31/10/2025 **Start Time: 02:00 pm**
Time Duration: 1.5 Hours **Max. Marks: 40**

Instructions:

1. Attempt all the questions sequentially/order; otherwise, you will be penalized.
2. Copying or exchanging any stationery/material is strictly not allowed in the exam.
3. Plagiarism will be thoroughly checked for all copies.
4. Assume no syntax error in all code snippets.

Section-A

1. Answer the following **[4+2=6]**
   a. Show the declaration of a pointer to an integer, say **ptr**. Suppose the above-defined pointer **ptr** points to an integer variable **a.** Then, write the C statements to the same and justify your answer with a proper argument
   b. Explain the difference between the following declarations:
      i. char x[ ] = {'2'};
      ii. char y[ ] = "2";

2. Explain a two-dimensional integer array declaration, say name is M, with column size 3 and row size 2. Define a function that adds the given integer, say n, to the 2nd column. Justify your answer with a suitable example. [**3+3=6]**

Section-B

3. Consider the following code segment. Write the output with justification. **[3]**

```
int i = 1, j = 5;
        while (i < j)
        {
           i = i + 2;
           j--;
        }
printf("%d\t%d", i, j);
```

4. What does fun(x) do?Assume any value of 'x'. Justify your answer with an example. **[5]**

```
int fun (long x)
{
```

```
    int n = 0;
    if (x)
       do {
          n++;
       } while (x &= (x – 1));
    return n;
}
```

5. What is the output of the following program? Justify your answer with an example**.** **[6]**

```
void fun() {
   int i;
   for (i = 1; i++ <= 5; printf("%d", i));
}
int main () {
   int j;
   for (j = 1; j++ <= 5; fun());
}
```

6. What is the output of this program with n=3? Justify your answer with the illustration. **[8]**

```
        void f(int n)
{
        if (n<=0)
        return;
        f(n-1);
        printf (“%d”, n);
        f(--n);
        f(n-1);
        printf (“%d”, n);
}
```

7. Consider the following two functions. **[6]**

void change (int x, int y)
{
x = x + y;
y = x – y;
x = x – y;
}
	void make (int * x, int * y)
	{
	*x = *x + *y;
	*y = *x – *y;
	*x = *x – *y;
	}

Explain at least two differences between the above two functions, change **()** and make **(),** with the respective function calls. Justify your answer with different inputs.